



RT-R Series

G2S Single-phase Energy Storage Inverter

KEY STRENGTHS

- Supports on/off-grid AC coupling function.
- Supports intelligent load management (customized).
- Compatible with three-phase applications and multi-unit parallel operation.
- Supports BMS remote upgrade function (customized).



G2S SERIES WITHOUT COLOR SCREEN



G2S COLOR SCREEN SERIES

Model	RT-R8KL1DA-G2S	RT-R10KL1DA-G2S
PV input data		
Max. PV input power (kW)	12	15
Max. PV input voltage (V)		550
Start-up voltage (V)		100
MPPT voltage range (V)		100~430
Rated PV input voltage (V)		360
No. of MPPT trackers		2
No. of strings per MPPT tracker		1 / 2
Max. input current per MPPT (A)		16 / 16+16
Max. short-circuit current per MPPT (A)		24 / 24 / 24

Model	RT-R8KL1DA-G2S		RT-R10KL1DA-G2S	
Battery input data				
Battery type	Lithium-ion / Lead-acid			
Max. charge / discharge power (kW)	8		10	
Battery voltage range (V)	40~58			
Rated battery voltage (V)	48			
Max. charging / discharging current (A)	160 / 160		200 / 200	
Rated charge / discharge current (A)	160 / 160		200 / 200	
Battery charging strategy	Self-adaption to BMS			
BMS communication	CAN			
AC output data (grid side)				
Rated output power (kW)	8		10	
Max. apparent output power (kVA)	8.8		10	
Max. apparent input power(kVA)	8.8		10	
Rated grid voltage (V)	230			
Grid voltage range (V)	176~270			
Grid input voltage (V)	230			
Rated grid frequency (Hz)	50 / 60			
Grid frequency range(Hz)	47~61			
Rated output current (A)	34.8		43.5	
Max. AC output current (A)	38.3		43.5	
Max. apparent input current (A)	38.3		43.5	
Power factor	>1 (0.8 leading ~ 0.8 lagging)			
Max. grid pass-through current (A)	43.5			
THDi	< 3%			
Grid type	L+N+PE			
AC output data (back-up)				
Rated output power (kW)	8		10	
Max. apparent output power (kVA)	8.8		10	
Rated output voltage (V)	230			
Rated output frequency (Hz)	50 / 60			
Rated output current (A)	34.8		43.5	
Max. AC output current (A)	38.3		43.5	
Peak output power	≥ 110%, 10mins; ≥120%, 1min; ≥ 130%, 1s; ≥ 150%, 100ms			
Back-up switch time (ms)	<20			
THDu	<3%			
Protection				
Supported protection	PV reverse polarity protection, Anti-islanding protection, Ground fault protection, Leakage current protection, Insulation resistance detection, Backup output short circuit protection, AC under-voltage protection, AC output over-current protection, AC over-voltage protection			
Surge protection	DC Type III / AC Type III			
Over voltage category	DC Type III / AC Type III			
Certifications and standards				
Certificates	ENIEC 61000-6-1:2019, EN IEC 61000-6-3:2021; EN 62109-1:2010; EN 62109-2:2011			
General data				
Europe efficiency	≥96.5%			
Ingress protection	IP66			
Operating temperature range (°C)	-25~+60			
Cooling	Forced air cooling			
Relative humidity	0~95% (non-condensing)			
Operating altitude (m)	0~2,000(> 2,000 power derating)			
Dimensions W*D*H (mm)	486×231×530			
Weight (kg)	25.8			
Inverter topology	Non-Isolated			
Noise emission (dB)	<55			
PV connection terminals	MC4			
Display and communication				
Display	LCD			
Communication	RS485 / CAN			
Connectivity	Wi-Fi and Lan (via external adapters)			